

BRATOLYUBOVKA, Russian Federation

WMO: 315210

Lat: 50.7863N

Lon: 129.4270E

Elev: 230

StdP: 98.59

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.5	(c) -34.2	(d) -38.9	(e) 0.1	(f) -36.3	(g) -36.9	(h) 0.1	(i) -34.0	(j) 5.4	(k) -18.9	(l) 4.8	(m) -20.2	(n) 1.1	(o) 110	(p) 0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.6	(c) 29.7	(d) 21.0	(e) 28.1	(f) 20.7	(g) 26.6	(h) 19.9	(i) 22.8	(j) 27.5	(k) 21.9	(l) 26.4	(m) 21.1	(n) 25.0	(o) 1.9	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.5	25.3	20.5	15.6	24.3	19.7	14.8	23.4	69.2	27.6	65.6	26.5	62.6	25.0	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.7	(b) 5.4	(c) 4.7	(e) -39.5	(f) 32.1	(g) 2.6	(h) 1.6	(i) -41.4	(j) 33.3	(k) -42.9	(l) 34.2	(m) -44.4	(n) 35.2	(o) -46.3	(p) 36.3		
(a) 6.7	(b) 5.4	(c) 4.7	(e) -39.7	(f) 24.1	(g) 2.6	(h) 1.2	(i) -41.6	(j) 24.9	(k) -43.1	(l) 25.6	(m) -44.6	(n) 26.2	(o) -46.5	(p) 27.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.1	-24.6	-19.2	-9.3	3.2	12.0	17.9	20.8	18.7	11.6	1.8	-12.1	-23.4		
	DBStd	16.95	4.96	5.43	6.39	5.24	4.02	3.59	2.41	3.01	4.56	5.08	6.99	5.42		
	HDD10.0	4718	1074	817	599	213	24	0	0	1	35	258	662	1037		
	HDD18.3	6905	1332	1050	857	453	201	51	6	30	205	514	912	1295		
	CDD10.0	1024	0	0	0	10	85	237	335	271	83	3	0	0		
	CDD18.3	169	0	0	0	1	3	38	83	42	3	0	0	0		
	CDH23.3	1394	0	0	0	9	79	409	556	308	33	0	0	0		
	CDH26.7	322	0	0	0	1	15	129	127	47	3	0	0	0		
(14) Wind	WSAvg	1.9	1.5	1.7	2.1	2.5	2.5	1.8	1.7	1.7	2.0	2.2	1.9	1.6		
(15) Precipitation	PrecAvg	554	7	7	11	23	50	82	123	109	65	32	19	12		
	PrecMax	794	20	27	32	56	132	214	250	230	146	72	41	34		
	PrecMin	385	2	0	0	1	10	25	25	43	12	9	5	2		
	PrecStd	102	4	6	10	14	30	42	58	45	36	17	9	8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-9.8	-2.0	8.2	24.5	29.0	33.0	31.7	29.6	26.7	18.7	7.0	-5.2		
		MCWB	-11.1	-4.4	2.5	12.0	16.0	23.2	22.1	21.7	18.3	11.0	3.1	-6.6		
	2%	DB	-12.6	-4.6	5.5	18.8	25.7	30.1	29.6	28.0	24.0	14.6	3.6	-9.8		
		MCWB	-13.5	-6.3	0.7	9.4	14.2	20.0	22.2	21.5	17.2	8.1	0.6	-10.5		
	5%	DB	-15.3	-7.6	3.2	15.0	23.0	27.9	28.0	26.6	22.0	12.1	0.6	-12.8		
		MCWB	-16.0	-9.2	-0.7	6.8	13.0	19.7	21.6	20.8	15.9	6.7	-1.3	-13.4		
	10%	DB	-17.3	-10.5	0.7	12.2	20.7	25.5	26.4	25.0	19.7	9.7	-1.5	-15.0		
		MCWB	-17.8	-11.7	-2.2	5.1	12.1	18.3	20.9	19.7	14.9	5.3	-3.1	-15.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-10.7	-4.3	2.9	12.6	17.5	24.2	24.1	23.9	19.6	11.7	3.6	-6.4		
		MCDB	-9.8	-2.8	7.7	23.7	24.9	31.7	28.9	27.8	24.2	17.7	6.6	-5.1		
	2%	WB	-13.4	-6.5	1.0	9.5	15.5	21.7	22.9	22.5	18.2	9.6	1.0	-10.4		
		MCDB	-12.6	-4.5	4.6	17.7	23.2	27.7	27.5	26.5	22.3	13.4	3.5	-9.9		
	5%	WB	-15.9	-9.1	-0.5	7.5	14.1	20.2	22.2	21.4	17.0	7.3	-1.0	-13.4		
		MCDB	-15.3	-7.6	2.8	13.8	21.0	26.0	26.8	25.1	20.8	11.2	0.8	-12.9		
	10%	WB	-17.8	-11.7	-2.0	5.7	13.0	19.1	21.5	20.5	15.6	5.7	-3.2	-15.6		
		MCDB	-17.3	-10.5	0.7	11.6	19.0	24.2	25.7	23.6	18.9	9.3	-1.6	-15.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.3	12.2	11.9	11.6	12.4	11.2	9.6	10.3	11.7	10.5	9.7	9.5		
		MCDBR	10.3	12.6	11.9	15.5	16.4	14.9	12.3	12.7	14.3	14.6	9.9	8.8		
		MCWBR	9.6	10.7	8.0	8.6	7.8	6.7	5.8	6.6	7.6	8.9	7.6	8.2		
	5% WB	MCDBR	10.3	12.4	11.0	14.5	14.1	12.9	10.2	10.2	11.9	12.5	9.7	8.7		
		MCWBR	9.6	10.6	7.5	8.1	7.5	6.5	5.3	5.6	6.7	8.2	7.7	8.2		
(40) Clear-Sky Solar Irradiance	taub		0.267	0.300	0.380	0.544	0.601	0.494	0.477	0.426	0.401	0.417	0.332	0.272		
	taud		2.138	2.058	1.884	1.682	1.654	2.012	2.112	2.246	2.223	1.996	2.073	2.090		
	Ebn at Noon		744	809	792	695	679	772	779	804	778	650	631	660		
	Edh at Noon		76	109	158	219	236	166	148	123	113	117	82	68		
(44) All-Sky Solar Radiation	RadAvg		1.63	2.86	4.44	5.04	5.44	5.95	5.52	4.82	3.80	2.48	1.60	1.26		
	RadStd		0.04	0.06	0.25	0.54	0.43	0.70	0.41	0.31	0.33	0.12	0.09	0.07		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page